# Understanding Inconsistent Values

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# What Are Inconsistent Values in Data?

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## Definition

Inconsistent values are data entries that deviate from the expected, valid, or standardized format. These values may not necessarily be missing or null — rather, they are **incorrect or contradictory** based on domain rules, logic, or context.

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## Examples

- A dataset about countries listing "UK", "U.K.", and "United Kingdom" in different rows.
- A column for gender that contains values like "Male", "male", "MALE", "M", "F", "femal", and "FEMALE".
- Product prices listed as "50$", "fifty", and "50.00 USD".

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## Why This Matters

- Distort analysis
- Cause incorrect statistical results
- Break groupings or aggregations
- Affect machine learning model accuracy

# Causes of Inconsistent Data Values

## 1 Human Entry Errors

Manual data entry often leads to typos, misspellings, and varied capitalization (e.g., "USA" vs. "Usa").

## 2 Lack of Standardization

Different systems, regions, or departments may use different naming conventions (e.g., "Product_ID" vs. "Product Number").

## 3 Merged Data from Multiple Sources

Combining datasets from different databases can lead to value inconsistencies due to format differences (e.g., "mm/dd/yyyy" vs. "dd-mm-yyyy").

## 4 Ambiguous Data Fields

Unclear instructions or field definitions can result in inconsistent interpretations (e.g., "Status" field with "Active", "1", "Yes", and "A" for the same state).

# Real-World Impacts of Inconsistent Values



## Inaccurate Reporting

When categories like "Bangladesh", "BD", and "B.D." are treated separately, regional analysis becomes unreliable.



## Broken Groupings in Dashboards

Business tools like Power BI or Excel may split sales by "Online" and "online" as separate categories.




## Poor Model Performance in AI/ML

Inconsistent labels or inputs degrade the quality of features, leading to inaccurate predictions.



## Increased Cleaning Cost

Analysts must spend more time cleaning and standardizing data, reducing time for actual analysis.

# Types of Inconsistent Values

### Format Inconsistencies

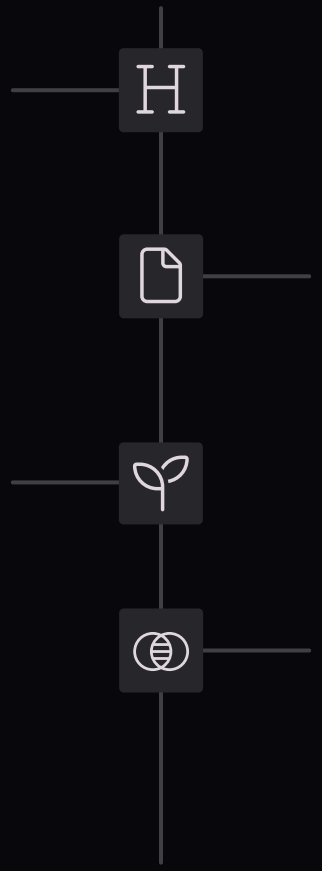
Same data type but different structures

E.g., Dates in "2023/01/01", "01-01-2023", and "Jan 1, 2023"

### Spelling and Case Errors

Typos, capital letters, or mixed case

E.g., "color", "Colour", "COLR"



### Semantic Inconsistencies

Different representations of the same value

E.g., "NY", "New York", and "N.Y."

### Logical Contradictions

Values that don't logically align

E.g., Age = 5 and Marital Status = Married

# Summary – Key Takeaways

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### Understanding Inconsistent Values

**Inconsistent values** refer to data entries that are incorrect or vary in format or meaning.

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### Root Causes

They arise due to manual entry, merging sources, and poor validation.

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### Business Impact

They negatively impact analysis, modeling, and decision-making.

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### Detection Methods

Detection involves profiling, logical checks, and aggregation tests.

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### Resolution Approaches

Fixing requires standardization, normalization, and cleaning routines.

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### Prevention Strategies

Prevention is best achieved through validation, standard rules, and automation.

# Thank You!

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